

Compressing Inter-Agent KV Relays in LatentMAS

Research report (technical, self-contained)

Audience: familiar with transformer internals (attention, KV cache, residual stream), new to this project

Primary backbone: Qwen3-14B ($L=40$, $d_{\text{model}}=5120$, GQA with $n_{\text{kv}}=8$, $d_h=128$).

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Abstract

LatentMAS runs three non-decoding agents that grow a shared KV cache (“relay”), then a Judger that decodes text from it. Empirically that relay is **content-insensitive within task** (wrong-question $\approx$ right-question) and therefore highly redundant. On GSM8K we reduce relay storage from ~ 100 MB/640 positions to ~ 2.5 MB/16 positions ($\approx 40\times$) at matched accuracy. Compression itself costs only ~ 20 ms per item (tensor ops over the cache)—negligible versus upstream latent forwards (~ 3 s) and more than repaid by faster Judger attention over a short prefix. Harder tasks (MATH, MedQA) remain compressible only at thicker residual budgets.

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1 System model

1.1 Pipeline

LatentMAS factorizes answering into four roles sharing one model:

$$x \xrightarrow{\text{Planner}} \xrightarrow{\text{Critic}} \xrightarrow{\text{Refiner}} C_{\text{relay}} \xrightarrow{\text{Judger}} y_{1:T}.$$

Upstream agents do **not** emit tokens. Agent a at latent step t computes

$$\begin{aligned} h_t^{(a)} &= f_{\theta}(W_a(h_{t-1}^{(a)}), C_{<t}), \\ C_t &= C_{t-1} \oplus \text{KV}(h_t^{(a)}), \end{aligned}$$

where W_a realigns last-layer hidden states into the embedding space and $\oplus$ appends new key/value slots. After K steps per upstream agent (plus role prefills), the Judger receives C_{relay} and runs ordinary autoregressive decoding.

Default regime here: $K=40$ latent steps, greedy Judger decode, generation budgets T_{max} task-dependent (GSM8K 768, MATH 2048, ...).

1.2 What “relay size” means

For a cache with L layers, n_{kv} KV heads, head dim d_h , and S positions, byte cost is

$$\text{Bytes}(C) = 2 L n_{\text{kv}} S d_h \cdot b,$$

with $b=2$ bytes for bf16/fp16. On Qwen3-14B ($L=40$, $n_{\text{kv}}=8$, $d_h=128$):

$$\text{MB}(S) \approx \frac{2 \cdot 40 \cdot 8 \cdot 128 \cdot S \cdot 2}{2^{20}} = \frac{S}{6.4} \text{ MB} \quad \Rightarrow \quad S=640 \mapsto \approx 100 \text{ MB}.$$

Measured GSM8K means: $S \approx 640$, ≈ 100.0 MB.

2 Prior negative result (why we compress)

Activation steering of upstream agents (SEAL / KV / learned CES contrasts) did **not** improve accuracy. Linear probes of upstream latents for final correctness are near-chance (AUCs ≈ 0.54 – 0.62), while Judger states are informative (≈ 0.69 – 0.80). So the relay is a poor place to “inject correctness.”

Mechanism controls instead show a **scaffold** picture:

Relay given to Judger	Typical finding
Real $C(x)$	Baseline acc / length
Shuffled $C(x')$, $x' \neq x$	Acc $\approx$ real (within a few pts on GSM8K/MATH)
Empty	Acc drops (esp. MedQA: $0.63 \rightarrow 0.27$)

Interpretation: within a task family, C supplies a structured, mostly **instance-agnostic** prior that shortens / stabilizes Judger decoding more than it transmits a lossless solution trace. High within-task redundancy $\Rightarrow$ aggressive position selection should be viable.

3 Compressor

3.1 Formal description

Let layer ℓ store $K^{(\ell)}, V^{(\ell)} \in \mathbb{R}^{1 \times n_{kv} \times S \times d_h}$. Choose budget B , sink s , optional rank r , importance σ .

Scores. Default `key_norm`, aggregated over heads so one index set is compatible with a dense HF `DynamicCache`:

$$\sigma_i = \sum_{h=1}^{n_{kv}} \|K_{h,i,:}^{(\ell)}\|_2, \quad i = 0, \dots, S-1.$$

Alternatives: value-norm, or recency $\sigma_i = i$.

Selection (plain-H / evict).

$$\mathcal{I} = \{0, \dots, s-1\} \cup \text{Top}_{B-s}(\{\sigma_i : i \geq s\}).$$

Sort $\mathcal{I}$ ascending (preserve causal order) and restrict

$$K' = K_{::,\mathcal{I},:}, \quad V' = V_{::,\mathcal{I},:}.$$

OBF backfill (obf). Let $\mathcal{E} = \{0, \dots, S-1\} \setminus \mathcal{I}$. Partition $\mathcal{E}$ (in original order) into r contiguous buckets and mean-pool each bucket to get synthetic slots $\tilde{K}, \tilde{V} \in \mathbb{R}^{1 \times n_{kv} \times r \times d_h}$. Append:

$$K_{\text{out}} = [K' \parallel \tilde{K}], \quad V_{\text{out}} = [V' \parallel \tilde{V}].$$

Output length: B (`evict`) or $B+r$ (`obf`). Default knobs in the headline GSM8K run: $B=16, s=4, r=8 \Rightarrow 16$ or 24 positions.

Complexity. One pass of norms + `topk` + `index_select` (+ cheap mean-pool):

$$\Theta(L n_{kv} S d_h) = \Theta(\text{numel}(C)),$$

i.e. linear in cache size—**not** another model forward.

Faithfulness caveat. Published H-OBF selects per-head via attention mass. We select **per layer** (shared indices across heads) so the result remains a standard dense cache consumable by **generate**. Scorer is pluggable.

3.2 Placement in the pipeline

$$C_{\text{relay}} \xrightarrow{\text{Compress}_{B,s,r}} \hat{C} \xrightarrow{\text{Judger} \pm \text{ASC}} y.$$

Upstream agents are always unsteered. Optional **ASC** = Judger-only SEAL residual at layer 28, coefficient 40, for length control.

4 Cost model — including compression overhead

End-to-end latency per item in the harness is

$$t_{e2e} = t_{\text{upstream}} + t_{\text{compress}} + t_{\text{decode}}.$$

Term	What it is	Scales with
t_{upstream}	$3(K+1)$ latent forwards + prefills	model FLOPs $\times K$
t_{compress}	norms / top- k / gather / optional pool	$\text{numel}(C)$
t_{decode}	Judger AR decode; attends over prefix $ \hat{C} $	$\approx \sum_t O(L n_h t_{\text{ctx}} d)$

4.1 Measured overhead (GSM8K, keep-16, seed 123, $n=60$)

From `report.json` means and paired contrasts:

Arm	Relay	Acc	Tok	t_{dec}	t_{e2e}	MB	Peak
A	full	0.817	529	2.84 s	5.87 s	100.0	36.8 GB
C	obf ($B+r=24$)	0.800	567	1.79 s	4.85 s	3.75	34.2 GB
E	evict ($B=16$)	0.817	562	1.78 s	4.82 s	2.50	34.1 GB

Paired contrasts ($C-A$):

- $\Delta t_{\text{decode}} \approx -1.05$ s (credible speedup)
- $\Delta t_{e2e} \approx -1.02$ s

Because t_{upstream} is shared and $t_{\text{compress}}^A = 0$,

$$\Delta t_{e2e} = t_{\text{compress}}^C + \Delta t_{\text{decode}} \Rightarrow t_{\text{compress}}^C \approx (-1.02) - (-1.05) \approx \mathbf{20-25} \text{ ms.}$$

So yes: **there is a compression cost**, but it is $\sim 0.7\%$ of t_{upstream} (≈ 3.0 s) and $\sim 1\%$ of baseline t_{e2e} . Net, compressed arms are **faster** end-to-end, not slower.

Why decode speeds up even when tokens rise (+37 mean tokens, C vs A): attention cost tracks **prefix length** S as well as generated length T . Cutting $S: 640 \rightarrow 16-24$ dominates a modest verbosity tax. ASC on the Judger removes that tax (Δ tokens $D-C \approx -58$, similar to $B-A$).

4.2 Memory accounting (important distinction)

Quantity	Full	Compressed @16	Notes
Stored relay after compress	100 MB	2.5 MB	headline 40×
Peak GPU during run	36.8 GB	34.1 GB	~ 2.6 GB drop
Upstream compute	paid in full	paid in full	compressor does not skip latent steps

Peak still includes model weights + activations while *building* the full cache; we compress **after** materialization. The 40× figure is **relay residency / Judger prefix memory**, not “whole process 40× cheaper.” A separate synthetic-scaffold line removes upstream compute; that is a different lever (saves FLOPs, historically not KV bytes when kept full-length).

4.3 When overhead would matter

t_{compress} only becomes comparable to decode if S is huge and T is tiny, or if you re-compress every decode step (we do not—once per item). Our scorer is $O(S)$ per layer; an attention-mass H-OBF port would cost more (needs attentions) and should be re-benchmarked.

Note: the harness records `compress_s` per row; aggregated `report.json` currently omits `mean_compress_s`. The 20–25 ms figure above is inferred from $\Delta t_{\text{e2e}} - \Delta t_{\text{decode}}$ and should match a direct mean of `compress_s`.

5 Experimental design ($2 \times 2 + \text{diagnostics}$)

Paired items; same upstream cache reused across Judger arms:

Arm	Relay	Judger ASC
A	full	off
B	full	on
C	obf	off
D	obf	on
E / F	evict $\pm$ ASC	optional
S	shuffled full	off
N	none	off

Primary contrasts:

$$\begin{aligned}
 \text{compression tax} &= C - A, \\
 \text{ASC} \mid \text{full} &= B - A, \\
 \text{ASC} \mid \text{obf} &= D - C, \\
 \text{interaction} &= (D - C) - (B - A).
 \end{aligned}$$

Target systems story: $\text{acc}(E) \approx \text{acc}(A)$, $\text{MB}(E) \ll \text{MB}(A)$, and if $T(C) > T(A)$ then ASC restores length (interaction ≈ 0 when both cells show the same ASC effect—observed on GSM8K tokens: interaction ≈ -1.9 tok, CI covers 0).

6 Results

6.1 GSM8K — iso-accuracy at $\approx 40\times$

Seed	Full acc	Evict@16	Obf@16	MB full→evict
42 (agg_b16)	0.817	0.817	0.800	100 → 2.5
123	0.817	0.817	0.800	100 → 2.5
7	0.817	0.817	0.800	100 → 2.5

Shuffled vs full (seed 123): 0.800 vs 0.817. Plain `evict` matches full accuracy; mean-pool rank-8 backfill is not necessary for this operating point (and uses $+r$ positions).

6.2 MATH — thicker residual

Full acc 0.783. Obf keep- B :

B	Acc	Approx MB ratio
128	0.800	$\sim 5\times$
64	0.733 (−5)	$\sim 9\times$
32	0.717 (−7)	$\sim 16\times$

6.3 MedQA — aggressive budgets break

Full acc 0.633. Obf:

B	Acc
128	0.567 (−7, soft)
64	0.517 (−12)
16–32	~ 0.35 – 0.37 (collapse)

None: 0.267. MedQA $\leftarrow$ GSM8K donor cache: 0.533 (−10)—cross-task transfer is not free.

6.4 Operating rule

$B^*(\text{task})$ increases with task difficulty / domain shift.

GSM8K admits $B=16$; MATH wants $B \sim 128$ for iso-acc; MedQA should not use $B \in \{16, 32\}$ under this scorer.

7 What is / is not claimed

Claimed.

1. Relay KV is within-task redundant; position selection yields large MB cuts at iso-acc on GSM8K (replicated).
2. Task-dependent compression frontier on MATH/MedQA.
3. Compressor wall-clock overhead is $O(\text{numel}(C))$ and empirically $\ll t_{\text{upstream}}, |\Delta t_{\text{decode}}|$.

Not claimed.

1. Elimination of upstream FLOPs (cache is still built once).
2. Universal $40\times$ on hard exams.
3. Byte-faithful per-head attention H-OBF.
4. AIME numbers from the long-decode cache-swap harness (truncation artifacts).

8 Notation

Symbol	Meaning
$C, \widehat{C}$	full / compressed relay KV
S, B, s, r	length / budget / sink / OBF rank
$t_{\text{upstream}}, t_{\text{compress}}, t_{\text{decode}}$	latency partition
ASC / SEAL	Judger-only activation steer for length
Arm A/C/E	full / obf / evict without ASC

Implementation: `seal/relay_compress.py`, `scripts/exp_2x2_relay_asc.py`.

Runbook: `docs/CACHE_COMPRESSION_PIPELINE.md`.

Full experiment log: `docs/MASTER_BRIEF.md`.